	#Size	BoolQ			CB			COPA			MultiRC (F1a)		
		FT	PT	PT-2	FT	PT	PT-2	FT	PT	PT-2	FT	PT	PT-2
$\text{BERT}_{\text{large}}$	335M	**77.7**	67.2	75.8	**94.6**	80.4	**94.6**	69.0	55.0	**73.0**	70.5	59.6	**70.6**
$\text{RoBERTa}_{\text{large}}$	355M	**86.9**	62.3	84.8	98.2	71.4	**100**	**94.0**	63.0	93.0	**85.7**	59.9	82.5
$\text{GLM}_{\text{xlarge}}$	2B	**88.3**	79.7	87.0	**96.4**	76.4	**96.4**	**93.0**	92.0	91.0	84.1	77.5	**84.4**
$\text{GLM}_{\text{xxlarge}}$	10B	88.7	**88.8**	**88.8**	**98.7**	98.2	96.4	**98.0**	**98.0**	**98.0**	**88.1**	86.1	**88.1**

	#Size	ReCoRD (F1)			RTE			WiC			WSC		
		FT	PT	PT-2	FT	PT	PT-2	FT	PT	PT-2	FT	PT	PT-2
$\text{BERT}_{\text{large}}$	335M	70.6	44.2	**72.8**	70.4	53.5	**78.3**	74.9	63.0	**75.1**	**68.3**	64.4	**68.3**
$\text{RoBERTa}_{\text{large}}$	355M	89.0	46.3	**89.3**	86.6	58.8	**89.5**	**75.6**	56.9	73.4	63.5	**64.4**	63.5
$\text{GLM}_{\text{xlarge}}$	2B	91.8	82.7	**91.9**	**90.3**	85.6	**90.3**	**74.1**	71.0	72.0	**95.2**	87.5	92.3
$\text{GLM}_{\text{xxlarge}}$	10B	**94.4**	87.8	92.5	**93.1**	89.9	**93.1**	**75.7**	71.8	74.0	**95.2**	94.2	93.3

Table 2: Results on SuperGLUE development set. P-tuning v2 surpasses P-tuning & Lester et al. (2021) on models smaller than 10B, matching the performance of fine-tuning across different model scales. (FT: fine-tuning; PT: Lester et al. (2021) & P-tuning; PT-2: P-tuning v2; **bold**: the best; underline: the second best).

	#Size	CoNLL03				OntoNotes 5.0				CoNLL04			
		FT	PT	PT-2	MPT-2	FT	PT	PT-2	MPT-2	FT	PT	PT-2	MPT-2
$\text{BERT}_{\text{large}}$	335M	**92.8**	81.9	90.2	91.0	**89.2**	74.6	86.4	86.3	85.6	73.6	84.5	**86.6**
$\text{RoBERTa}_{\text{large}}$	355M	92.6	86.1	**92.8**	**92.8**	**89.8**	80.8	**89.8**	**89.8**	88.8	76.2	88.4	**90.6**
$\text{DeBERTa}_{\text{xlarge}}$	750M	**93.1**	90.2	**93.1**	**93.1**	90.4	85.1	90.4	**90.5**	89.1	82.4	86.5	**90.1**

	#Size	SQuAD 1.1 dev (EM / F1)								SQuAD 2.0 dev (EM / F1)							
		FT		PT		PT-2		MPT-2		FT		PT		PT-2		MPT-2	
$\text{BERT}_{\text{large}}$	335M	**84.2**	**91.1**	1.0	8.5	77.8	86.0	82.3	89.6	**78.7**	**81.9**	50.2	50.2	69.7	73.5	72.7	75.9
$\text{RoBERTa}_{\text{large}}$	355M	**88.9**	**94.6**	1.2	12.0	88.5	94.4	88.0	94.1	**86.5**	**89.4**	50.2	50.2	82.1	85.5	83.4	86.7
$\text{DeBERTa}_{\text{xlarge}}$	750M	90.1	95.5	2.4	19.0	**90.4**	**95.7**	89.6	95.4	88.3	91.1	50.2	50.2	**88.4**	**91.1**	88.1	90.8

	#Size	CoNLL12				CoNLL05 WSJ				CoNLL05 Brown			
		FT	PT	PT-2	MPT-2	FT	PT	PT-2	MPT-2	FT	PT	PT-2	MPT-2
$\text{BERT}_{\text{large}}$	335M	84.9	64.5.	83.2	**85.1**	**88.5**	76.0	86.3	**88.5**	82.7	70.0	80.7	**83.1**
$\text{RoBERTa}_{\text{large}}$	355M	**86.5**	67.2	84.6	86.2	**90.2**	76.8	89.2	90.0	85.6	70.7	84.3	**85.7**
$\text{DeBERTa}_{\text{xlarge}}$	750M	86.5	74.1	85.7	**87.1**	**91.2**	82.3	90.6	**91.2**	86.9	77.7	86.3	**87.0**

Table 3: Results on Named Entity Recognition (NER), Question Answering (Extractive QA), and Semantic Role Labeling (SRL). All metrics in NER and SRL are micro-f1 score. (FT: fine-tuning; PT: P-tuning & Lester et al. (2021); PT-2: P-tuning v2; MPT-2: Multi-task P-tuning v2; **bold**: the best; underline: the second best).

Ratios of task-specific parameters (e.g., 0.1%) are derived from comparing continuous prompts' parameters with transformers' parameters. Another thing to notice is that our experiments are all conducted in the fully-supervised setting rather than few-shot setting.

NLU Tasks. First, we include datasets from SuperGLUE (Wang et al., 2019) to test P-tuning v2's general NLU ability. Additionally, we introduce a suite of sequence labeling tasks, including named entity recognition (Sang and De Meulder, 2003; Weischedel et al., 2013; Carreras and Màrquez, 2004), extractive Question Answering (Rajpurkar et al., 2016), and semantic role labeling (Carreras and Màrquez, 2005; Pradhan et al., 2012)).

Pre-trained Models. We include BERT-large (Devlin et al., 2018), RoBERTa-large (Liu et al., 2019), DeBERTa-xlarge (He et al., 2020), GLM-xlarge/xxlarge (Du et al., 2021) for evaluation. They are all bidirectional models designed for NLU tasks, covering a wide range of sizes from about 300M to 10B.

Multitask Learning. For the multi-task setting, we combine the training sets of the datasets in each task type (e.g., combing all training sets of semantic role labeling). We use separate linear classifiers for each dataset while sharing the continuous prompts (Cf. Appendix A).

	SST-2	RTE	BoolQ	CB
CLS & linear head	96.3	88.4	84.8	96.4
Verbalizer & LM head	95.8	86.6	84.6	94.6

Table 4: Comparison between [CLS] label with linear head and verbalizer with LM head on RoBERTa-large.

4.1 P-tuning v2: Across Scales

Table 2 presents P-tuning v2's performances across model scales. In SuperGLUE, performances of Lester et al. (2021) and P-tuning at smaller scales can be quite poor. On the contrary, P-tuning v2 matches the fine-tuning performance in all the tasks at a smaller scale. P-tuning v2 even significantly outperforms fine-tuning on RTE.

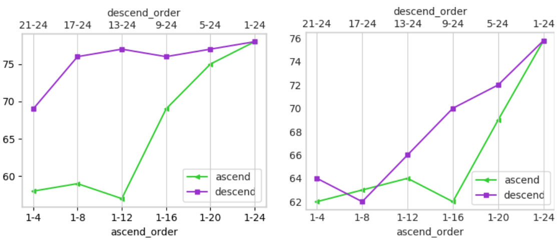
In terms of larger scales (2B to 10B) with GLM (Du et al., 2021), the gap between Lester et al. (2021); Liu et al. (2021) and fine-tuning is gradually narrowed down. On 10B scale, we have a similar observation as Lester et al. (2021) reports, that prompt tuning becomes competitive to fine-tuning. That said, P-tuning v2 is always comparable to fine-tuning at all scales but with only 0.1% task-specific parameters needed comparing to fine-tuning.

4.2 P-tuning v2: Across Tasks

From Table 3, we observe that P-tuning v2 can be generally comparable to fine-tuning on all tasks. P-tuning and Lester et al. (2021) show much poorer performance, especially on QA, which might be the most challenging of the three tasks. We also notice that there are some abnormal results of Lester et al. (2021) and P-tuning on SQuAD 2.0. This is probably because SQuAD 2.0 contains unanswerable questions, which causes optimization challenges for single-layer prompt tuning. Multi-task learning generally brings significant improvements to P-Tuning v2 over most tasks except for QA.

4.3 Ablation Study

Verbalizer with LM head v.s. [CLS] label with linear head. Verbalizer with LM head has been a central component in previous prompt tuning approaches. However, for P-tuning v2 in a supervised setting, it is affordable to tune a linear head with about several thousand parameters. We present our comparison in Table 4, where we keep other hyper-parameters and only change [CLS] label with linear head to verbalizer with LM head. Here, for simplicity, we use "true" and "false" for SST-2, RTE and BoolQ; "true", "false" and "neutral" for CB. Results indicate that there is no significant difference between performances of verbalizer and [CLS].



(a) RTE (b) BoolQ

Figure 3: Ablation study on prompt depth using BERT-large. "[x-y]" refers to the layer-interval we add continuous prompts (e.g., "21-24" means we are add prompts to transformer layers from 21 to 24). Same amount of continuous prompts added to deeper transformer layers (i.e., more close to the output layer) can yield a better performance than those added to beginning layers.

Prompt depth. The main difference between Lester et al. (2021); (Liu et al., 2021) and P-tuning v2 is the multi-layer continuous prompts. To verify its exact influence, given a certain number of k layers to add prompts, we select them in both ascending and descending order to add prompts; for the rest layers, we left them untouched. As shown in Figure 3, with the same amount of parameters (i.e., num of transformer layers to add prompts), adding them in the descending order is always better than in the ascending order. In the RTE case, only adding prompts to layers 17-24 can yield a very close performance to all layers.

5 Conclusions

We present P-tuning v2, a prompt tuning method. Despite its relatively limited technical novelty, it contributes to a novel finding that prompt tuning can be comparable to fine-tuning universally across scales (from 330M to 10B parameters) and tasks. With high accuracy and parameter efficiency, P-Tuning v2 can be a potential alternative for fine-tuning and a strong baseline for future work.

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References

Tom B Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*.

Xavier Carreras and Lluís Màrquez. 2004. Introduction to the CoNLL-2004 shared task: Semantic role labeling. In *Proceedings of the Eighth Conference on Computational Natural Language Learning (CoNLL-2004) at HLT-NAACL 2004*, pages 89–97, Boston, Massachusetts, USA. Association for Computational Linguistics.

Xavier Carreras and Lluís Màrquez. 2005. Introduction to the CoNLL-2005 shared task: Semantic role labeling. In *Proceedings of the Ninth Conference on Computational Natural Language Learning (CoNLL-2005)*, pages 152–164, Ann Arbor, Michigan. Association for Computational Linguistics.

Xiang Chen, Xin Xie, Ningyu Zhang, Jiahuan Yan, Shumin Deng, Chuanqi Tan, Fei Huang, Luo Si, and Huajun Chen. 2021. Adaprompt: Adaptive prompt-based finetuning for relation extraction. *arXiv preprint arXiv:2104.07650*.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv e-prints*.

Zhengxiao Du, Yujie Qian, Xiao Liu, Ming Ding, Jiezhong Qiu, Zhilin Yang, and Jie Tang. 2021. All nlp tasks are generation tasks: A general pretraining framework. *arXiv preprint arXiv:2103.10360*.

Tianyu Gao, Adam Fisch, and Danqi Chen. 2020. Making pre-trained language models better few-shot learners. *arXiv preprint arXiv:2012.15723*.

Yuxian Gu, Xu Han, Zhiyuan Liu, and Minlie Huang. 2021. Ppt: Pre-trained prompt tuning for few-shot learning. *arXiv preprint arXiv:2109.04332*.

Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2020. Deberta: Decoding-enhanced bert with disentangled attention.

Boseop Kim, HyoungSeok Kim, Sang-Woo Lee, Gichang Lee, Donghyun Kwak, Dong Hyeon Jeon, Sunghyun Park, Sungju Kim, Seonhoon Kim, Dongpil Seo, et al. 2021. What changes can large-scale language models bring? intensive study on hyperclova: Billions-scale korean generative pretrained transformers. *arXiv preprint arXiv:2109.04650*.

Brian Lester, Rami Al-Rfou, and Noah Constant. 2021. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691*.

Xiang Lisa Li and Percy Liang. 2021. Prefix-tuning: Optimizing continuous prompts for generation. *arXiv preprint arXiv:2101.00190*.

Xiao Liu, Yanan Zheng, Zhengxiao Du, Ming Ding, Yujie Qian, Zhilin Yang, and Jie Tang. 2021. Gpt understands, too. *arXiv:2103.10385*.

Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv e-prints*.

Sewon Min, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2021. Noisy channel language model prompting for few-shot text classification. *arXiv preprint arXiv:2108.04106*.

Sameer Pradhan, Alessandro Moschitti, Nianwen Xue, Olga Uryupina, and Yuchen Zhang. 2012. CoNLL-2012 shared task: Modeling multilingual unrestricted coreference in OntoNotes. In *Joint Conference on EMNLP and CoNLL - Shared Task*, pages 1–40, Jeju Island, Korea. Association for Computational Linguistics.

Guanghui Qin and Jason Eisner. 2021. Learning how to ask: Querying lms with mixtures of soft prompts. *arXiv preprint arXiv:2104.06599*.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. 2019. Exploring the limits of transfer learning with a unified text-to-text transformer. *arXiv preprint arXiv:1910.10683*.

Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. Squad: 100,000+ questions for machine comprehension of text.

Erik F Sang and Fien De Meulder. 2003. Introduction to the conll-2003 shared task: Language-independent named entity recognition. *arXiv preprint cs/0306050*.

Timo Schick and Hinrich Schütze. 2020. It's not just size that matters: Small language models are also few-shot learners. *arXiv preprint arXiv:2009.07118*.

Taylor Shin, Yasaman Razeghi, Robert L Logan IV, Eric Wallace, and Sameer Singh. 2020. Autoprompt: Eliciting knowledge from language models with automatically generated prompts. *arXiv preprint arXiv:2010.15980*.

Alex Wang, Yada Pruksachatkun, Nikita Nangia, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2019. SuperGLUE: